Steve Wong

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Passionate about full-stack software development and user-centered, interactive applications, with a strong foundation in programming and a proven ability to contribute to impactful projects.

EDUCATION

- **University of Illinois Urbana-Champaign (UIUC), Urbana, IL**
    - B.S. in Computer Science (Grainger College of Engineering) | Expected Graduation: May 2030
- **Dougherty Valley High School**, San Ramon, CA - High School Diploma: June 2026
- Diablo Valley College/Compton College Dual Enrollment

ACADEMIC COURSES | SKILLS

- Languages:
    - Python, C++, Java, JavaScript, TypeScript, x86 Assembly, SQL
- Frameworks/Libraries:
    - Git, Firebase, NumPy, Pandas, React, Node.js, SQL, PyTorch, OpenCV, YOLO
- Coursework:
    - Multivariable Calculus, Linear Algebra, Differential Equations, Data Structures, Advanced C/C++

TECHNICAL PROJECTS

Swimmer Split Tracker | PyTorch, YOLO, React, Supabase

- Architected a highly scalable web application to track multi-swimmer lap splits, optimizing front-end state management for simultaneous data inputs.
- Initially engineered a PyTorch/YOLO computer vision pipeline to track swimmer movements/splits, but executed an architectural pivot to a state-managed web timer to overcome real-world environmental noise (water refraction overlapping athletes) to better support the swim coach.

Hydrio | Telemetry Web Application | Python, OpenCV, Firebase Realtime Database

- Built an asynchronous data pipeline connecting a custom-trained Python-based computer vision model, NoSQL Firebase backend, and a UI centered frontend website.
- Optimized query paths to handle high-frequency sensor updates, enabling the frontend to dynamically render incoming CV data without latency bottlenecks

Computer Vision Boot Camp | Founder & Lead Organizer

- Organized and led boot camps teaching fundamental Computer Vision concepts to middle and high school students

PROFESSIONAL EXPERIENCES

Readyfly | Data Analyst Intern (November 2023 – Present)

- Designed and implemented data cleanup and sorting algorithms for a large dataset of job listings.
- Developed a database schema and queries to organize and retrieve job listing data.
- Automated data processing and visualization using Python scripts.
- Collaborated with team members to integrate data analysis into an AI Job report.

NASA / George Mason University | Data Science Research Intern (June 2025 – August 2025)

- Analyzed massive TESS light curve datasets using AstroImageJ, engineering data visualizations to verify photometric exoplanet transits.
- Processed complex, noisy time-series data to extract key signals, successfully identifying and verifying rare exoplanetary events for ongoing astronomical research.

UCLA | Student Researcher @COSMOS (July 2025 – August 2025)

- Modeled rat neural spatial navigation by applying single and multi-layer perceptrons to decode complex biological place-cell signals.
- Collaborated within a brain-inspired computing cohort to analyze direction-sensing data, bridging biological neural processes with machine learning architectures.